Improving Synchronization and Channel Estimation Robustness for OFDM-Based Integrated Sensing and Communication in GNU Radio

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M3 Report, November 1, 2026

1 Summary of M3 progress

This report covers pull request gnuradio/gnuradio#8233, a Required option for the GRC Parameter block. It gives the status of the pull request, the changes and why they matter, the test results so far, the maintainer feedback and how I responded, a revised SOTA positioning, and the plan for the next contribution. The GSoC 2027 proposal is a separate document.

Since M2, I updated the pull request description to follow the GNU Radio template after a maintainer asked for it, and I tested the Qt GUI workflow, which passed. The no-GUI workflow passed all its checks in M2. The first Bokeh GUI test run showed a failure, and I am tracing the cause. The next contribution, the gr-digital and gr-radar change for ISAC, builds on the literature review and gap analysis from M1.

Table 1: Contribution status.

Item	Status
Contribution	Required option for the GRC Parameter block
Pull request	github.com/gnuradio/gnuradio/pull/8233, opened October 11, 2026
Related issue	#5096, "GRC: Allow for required parameters", open since September 24, 2021
State	Draft. A maintainer converted it to a draft on October 15, 2026 and added the label "Needs Action". The description now follows the GNU Radio template.
Size	4 files changed, 30 insertions, no deletions; one commit, f7afafee6, signed off
Testing	No-GUI and Qt GUI workflows passed. Bokeh GUI first run is being traced
Checks	DCO passed. One build run failed because it checked the wrong file; I am tracing it
Reviews	One maintainer comment, about the template. No code review yet
Next contribution	gr-digital and gr-radar change for ISAC (Section 6)

2 The contribution

The change adds a Required setting to the GRC Parameter block. When it is set to Required, the generated Python script defines the matching command-line argument with required=True.

Otherwise the argument keeps its configured default. It consists of a new field in grc/blocks/parameter.block.yml and the same two-branch logic in three Mako workflow templates (no GUI, Qt GUI, Bokeh GUI). The default is Optional, so existing flowgraphs generate the same code as before.

```
% if required_ == 'True':
parser.add_argument(
    ${ ', '.join(switches) }, dest="${param.name}", type=${type_}, required=True,
    help="Set ${param.params['label'].get_evaluated() or param.name}")
% else:
parser.add_argument(
    ${ ', '.join(switches) }, dest="${param.name}", type=${type_}, default=${ default },
    help="Set ${param.params['label'].get_evaluated() or param.name} [default=%(default)r]")
% endif
```

The code in the commit is unchanged since M2. The work in M3 went into the pull request description, the Qt GUI test and the Bokeh GUI test.

3 Technical significance

Issue #5096 asks for a way to mark a GRC parameter as required so the generated Python has no default, for example for a file name the user must supply. It is labeled Feature Request, GRC, codegen and good first issue, and has been open since 2021. Contributors asked to work on it in March 2023 and March 2026, so the pull request answers a request several people have made.

A flowgraph can now refuse to start without an input it cannot guess. The block definition only controls the properties dialog, so the behavior had to be implemented in the code generator, where the templates decide what Python is written. The optional path generates the same code as before, which keeps the review small: 30 added lines and no deletions.

Because the issue has had other interested contributors, the maintainers may prefer a different implementation. I will ask for direction in the pull request.

4 Testing and validation

4.1 No-GUI workflow

I generated a script from a flowgraph with two integer Parameter blocks, required_count (Required, value 10) and optional_count (Optional, default 20), using the source-tree workflow and block definitions instead of the installed GNU Radio 3.10.12.0. The generated code had required=True for the first and default=20 for the second. Running the script without --required-count printed an error and returned exit code 2. Running it with --required-count 5 passed argument parsing. The help text showed the required argument without brackets and the optional one with its default. py_compile and git diff --check passed.

```
parser.add_argument(
    "--required-count", dest="required_count", type=intx, required=True,
    help="Set Required count")

parser.add_argument(
    "--optional-count", dest="optional_count", type=intx, default=20,
    help="Set Optional count [default=%(default)r]")
```

4.2 Qt GUI workflow

I generated the same flowgraph with the Qt GUI workflow template and ran the checks on it. The Qt GUI workflow passed. This confirms that the second of the three templates produces the same required and optional behavior as the no-GUI one.

4.3 Bokeh GUI workflow

The first Bokeh GUI test run showed a failure. I am tracing whether the cause is in the Bokeh template or in the test setup, and I will rerun the test once it is fixed.

4.4 Continuous integration

The DCO check passed ("All commits are signed off!"). Of the build runs, one failed because it checked the wrong file. My reading is that the failure comes from that check and not from the change, and I will confirm it in the run log and rerun.

Table 2: Test results so far.

Check	Result	Notes
No-GUI: required argument omitted	Passed	Parser error, exit code 2
No-GUI: required argument supplied	Passed	Argument parsing accepted the value
No-GUI: optional default	Passed	default=20 in code and help text
No-GUI: help message	Passed	Required and optional shown correctly
Python syntax, git diff --check	Passed	No errors, no output
Qt GUI workflow	Passed	Same checks as no-GUI
DCO	Passed	All commits signed off
Bokeh GUI workflow, first run	Failure found	Cause being traced; rerun planned
Build run on wrong file	Failure found	Not caused by the change; to be confirmed in the log

4.5 Backward compatibility

Block.import_data() in grc/core/blocks/block.py skips saved parameter keys that the current block definition does not contain:

```
for key, value in parameters.items():
    try:
        self.params[key].set_value(value)
    except KeyError:
        continue
```

An older flowgraph has no required key, so the new field keeps its declared default, False, and the parameter stays optional. I plan to add a regression test that loads an older flowgraph without the field.

5 Maintainer feedback and my response

On October 15, 2026 the maintainer mbr0wn commented that the GNU Radio pull request template should not be bypassed, because the checks in it help the maintainers see that the work was done properly. They converted the pull request to a draft so I could fix it and added the label "Needs Action". There has been no review of the code itself.

Table 3: Feedback and response.

Feedback	Response	Status
Pull request description did not follow the template (mbr0wn, October 15)	Updated the description to follow the GNU Radio template. The change was made directly in the description	Done
Draft with label "Needs Action"	The requested change is made. Mark ready for review once the Bokeh test and the failed build run are sorted	In progress
Build run failure	Trace the failure in the log, confirm it is the wrong file, rerun	In progress

The template puts testing, scope and sign-off in a fixed place for reviewers. For the next pull request I will read the contribution guidelines and fill in the template before opening it.

6 Next contribution: ISAC in gr-digital and gr-radar

The GRC pull request gave me a complete run through the GNU Radio contribution process: a branch, tests, a DCO sign-off, a pull request linked to an issue, CI, and maintainer feedback. It does not touch frame timing, synchronization or channel estimation, so it is separate from the ISAC problem defined in M1.

The ISAC contribution is the next one. The groundwork is in place from M1: the literature review, the table of the gap, and the finding that gr-digital and gr-radar do not exchange synchronization or channel information. The plan is to choose one scoped change, discuss the design with maintainers before writing code, implement it with unit and integration tests, and evaluate it with synchronization error, channel-estimation error (NMSE), BER and range error across SNR and delay. The GSoC 2027 proposal lays this out in phases.

7 Revised SOTA positioning

Table 4 repeats the M1 comparison and adds a row for pull request #8233. The new row marks every ISAC column as No, because the GRC change does not address them.

Table 4: Related work, the GRC pull request and the proposed ISAC project.

Work	OFDM	GNU Radio impl.	Sync/CFO robustness	Radar range / channel est.	Joint comm + radar eval.
Sturm & Wiesbeck (2011)	Yes	No	No	Yes	No
Liu et al. (2020)	Survey	No	No	Survey	No
Liu et al. (2022)	Survey	No	No	Survey	No
Dai et al. (2025)	Yes	No	No	Yes	Partial
Vaezi et al. (2026)	Partial	No	No	Partial	Partial
Qaisar et al. (2026)	No	No	No	No	No
Doly et al. (2023)	Yes	Yes	No	Yes	Partial
KIT-CEL, gr-radar	Yes	Yes	No	Yes	No
This work: GRC required parameter (PR #8233)	No	Yes	No	No	No
Proposed ISAC project	Yes	Yes	Yes	Yes	Yes

In the GNU Radio ecosystem, the OFDM blocks in gr-digital and the OFDM radar blocks in gr-radar were built under different assumptions about timing, and they do not exchange synchronization or channel information. The papers I reviewed cover theory, surveys and one GNU Radio demonstration (Doly et al.). None measures communication and radar performance together on one shared estimate, and that is the gap the proposed project targets.

The GRC contribution improves tooling for building flowgraphs. For the ISAC project it supplies practical experience with the repository, the pull request template, DCO sign-off and CI, which the gr-digital and gr-radar work will use.

8 Next steps

- Finish pull request #8233: fix and rerun the Bokeh GUI test, trace the failed build run and rerun it, then mark the pull request ready for review.
- Run the GRC tests with pytest and add a regression test for older flowgraphs.
- Select the first scoped ISAC change, post the design to the GNU Radio community, and implement it with tests.
- Evaluate the change with the four metrics and prepare the final report, repository and video, following the M1 timeline.
- Revise the GSoC 2027 proposal after feedback from the community.

References

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3. F. Liu et al., "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," IEEE J. Sel. Areas Commun., vol. 40, no. 6, pp. 1728-1767, 2022.

4. S. A. Doly, A. Chiriyath, W. Standage-Beier, and D. W. Bliss, "GNU Radio realization of Waveform Co-design for Joint Radar-Communications system using SDRs," Proceedings of the GNU Radio Conference, vol. 8, no. 1, 2023.

5. Communications Engineering Lab, Karlsruhe Institute of Technology, "gr-radar," GitHub, https://github.com/kit-cel/gr-radar

6. Q. Dai et al., "A Tutorial on MIMO-OFDM ISAC: From Far-Field to Near-Field," arXiv:2504.19091, 2025.

7. M. Vaezi, G. Amarasuriya Aruma Baduge, E. Ollila, and S. A. Vorobyov, "A Tutorial on AI-Empowered Integrated Sensing and Communications," IEEE Communications Surveys & Tutorials, vol. 28, 2026.

8. M. U. F. Qaisar et al., "The Role of ISAC in 6G Networks: Enabling Next-Generation Wireless Systems," IEEE Trans. Network Science and Engineering, vol. 13, 2026.

9. GNU Radio pull request #8233, https://github.com/gnuradio/gnuradio/pull/8233, and issue #5096, https://github.com/gnuradio/gnuradio/issues/5096.